NETFLOW

L I Q U I D L E V E L M E A S U R E M E N T · M B L S E R I E S

MAGNETIC LEVEL INDICATOR

By-pass type analog level indicator for tanks and vessels, with no contact to the medium

The magnetic by-pass level indicator allows the liquid level in tanks to be monitored from outside by means of a magnetic float and a moving colour-changing indicator.

PN10
10 BAR

100 °C / 250 °C
MAX. OPERATING TEMPERATURE

5800 mm
ONE-PIECE LENGTH

AISI 304 / 316
FULLY STAINLESS STEEL



MBL STANDART

- Indicator profile with no contact to the medium
- Analog indicator easily readable from a distance
- Robust, shock-resistant body
- Dry contacts at any point with limit switches
- Easy installation with standard and swivel flanges
- Easy maintenance with the valved model
- Custom manufacturing for different connections and lengths
- Level reading in percent, volume or metres with the scaled model

TECHNICAL SPECIFICATIONS

Pressure (Pmax)	PN10
Temperature (Tmax)	100 °C · optional max. 250 °C
Connection material	Carbon steel St 37.2 · stainless steel (optional)
Connection types	Threaded G1/2" · G3/4" · G1" Flanged DN15 · DN20 · DN25
Max. length	5800 mm, one piece
Float	AISI 316 · Ø34 mm
Tube	AISI 304 (AISI 316 optional) · Ø38 mm
Indicator profile	Aluminium
Flaps	ABS plastic · optional glass fibre (200 °C resistant)
Contact / signal	Dry contact via limit switch · 0–10 V / 4–20 mA (MBL 30)

DESIGN, OPERATING PRINCIPLE AND DIMENSIONS

1. **Communicating vessels.** The indicator is connected to the tank as a by-pass via two process connections. The liquid level in the body tube is the same as in the tank.
2. **Magnetic float.** The float inside the tube floats on the liquid surface and moves with the level.
3. **Flap indicator.** The magnet in the float rotates the flaps in the profile by 180°. Below the level they show red, above white.
4. **Contact and signal.** Limit switches mounted on the profile provide dry contacts at the required points. In MBL 30 models the level is transmitted as an analog signal.

MODELS

MBL STANDARD

Mounted on the tank as a by-pass. Level changes are monitored visually via the flaps. Limit switches can be fitted.

MBL 30

The level signal is transmitted to a digital controller, which displays the level in percent, volume, mass or metres.

MBL 30 DIRECT OUTPUT

The level transmitter is integrated into the indicator. The level is sent directly as a 0–10 V or 4–20 mA signal.

ACCESSORIES

ACCESSORY	DESCRIPTION	CONTACT / OUTPUT	VOLTAGE	TEMP.
Limit switch	Mounted on the indicator profile, provides a dry contact at the required point.	1.5 A · 50 W max. NO·NC	250 VAC · 200 VDC	-60…+125 °C
High-temp switch	Mounted on the profile in high-temperature applications.	1.5 A · 50 W max. NO·NC	250 VAC · 200 VDC	-60…+250 °C
Controller	Switches between the limit switches and the devices to be controlled.	6 contact inputs · 3×10 A NO+NC	220 VAC 50 Hz	--
Digital controller	Used with MBL 11. The level is shown on the display.	3×10 A NO+NC · 4–20 mA · 0–10 V	220 VAC 50 Hz	--
Analog indicator	Connected to MBL 11 or a digital controller.	--	24 V	--

INSTALLATION AND USE

01 The indicator is mounted vertically on two process connections on the tank side wall.

02 The centre distance (L) is specified in the order; manufactured in one piece up to 5800 mm.

03 With the valved model, the indicator can be serviced without draining the tank.

04 Limit switches are fixed at the required level on the indicator profile.

05 Sediment and residue in the tube are drained periodically through the drain plug.

CAUTION: There must be no strong magnetic field or magnetic material near the indicator.

Custom manufacturing for other connections, lengths, pressures and temperatures. Orders and technical support: NET FLOW · netflw.com · info@netflw.com · +90 553 296 52 42